

Project 1.

Stage 1: Problem Identification.

1.1 Description of the problem.

A businessman Dr Chanakira Trevor wants to determine how much his investment of \$3000 will grow if it increases by 20% each month over a period of 4 years.

1.2 Statement of Intent.

This project aims to use the principles of geometric progression to predict the total value of capital after 4 years of monthly 20% growth.

1.3 Project Specifications.

- Initial Capital : \$3000
- Monthly Increase Rate : 20% (0.20)
- Period 4 years = 48 months.
- formulae $T_n = ar^{n-1}$ or total $S_n = \frac{a(1-r^n)}{1-r}$

Stage 2: Investigation of Related Ideas.

2.1 Related Ideas.

- 1) Simple Interest - fixed addition per period (not suitable here)
- 2) Arithmetic Progression (AP) - equal additions, not percentage based

3 Geometric Progression (GP) - Multiplicative Increase, fits the monthly 20% growth scenario.

2.2 Merits and Demerits.

Method	Merits	Demerits.
Simple Interest	Easy to calculate	Doesn't account for Compound growth.
Arithmetic Progression	Suitable for linear growth	Doesn't apply to percentage increase.
Geometric Progression	Accurately models Compound percentage Increase	Involves large powers, May require Calculator.

Stage 3: Development of chosen Idea.

3.1 Indication of choice.

Geometric Progression is chosen due to the percentage-based monthly growth.

3.2 Justification.

Growth is multiplicative, increasing each month by 10%.

- GP models this perfectly.

3.3 Refinements.

- 1) let $a = 3000$, initial capital.
- 2) $r = 1 + 0,20 = 1,20$, monthly growth factor
- 3) $n = 48$, months in 4 years.
- 4) final amount after 4 years: final calculation shown clearly with step by step use of the GP formula.

Stage 4 Presentation of final solution.

4.1 Calculations.

- tables showing growth every 6 months.
- formula used: $T_n = ar^{n-1}$
- final capital after 4 years.

$$T_n = ar^{n-1}$$

$$a = 3000$$

$$r = 1,20$$

$$n = 48$$

$$T_n = 3000 (1,20)^{48-1}$$

$$= 3000 (1,20)^{47}$$

$$= \$15\,799\,371,79$$

Month 0.

Capital in USD\$ = 3000.

Month 12.

$$\begin{aligned} T_{12} &= ar^{n-1} \\ &= 3000(1,20)^{12-1} \\ &= 22\,2960,25. \end{aligned}$$

Capital in USD\$ = 22 2960,25.

Month 24

$$\begin{aligned} T_{24} &= ar^{n-1} \\ &= 3000(1,20)^{24-1} \\ &= 19\,8742,12. \end{aligned}$$

Month 36.

$$\begin{aligned} T_{36} &= ar^{n-1} \\ &= 3000(1,2)^{36-1} \\ &= 1\,772\,004,69. \end{aligned}$$

Table of month vs capital

Month	Capital / USD\$.
0	3000
12	22 2960,25.
24	19 8742,12
36	1 772 004,69
48	15 799 371,79.

Stage 5: Evaluation and Recommendations.

5.1 Reference to Statement of Intent.

The project successfully predicts the capital using geometric progression, fulfilling the aim.

5.2 Challenges faced.

- Difficulty calculating high powers manually
- Interpreting exponential growth initially.

5.3 Recommendations.

- Use Software tools like Excell or Calculators for accuracy.
- Teach business owners about Compound growth for financial planning.

Stage 6: References.

- New General Mathematics: O level Book 3.
- Schaum's outlines: Business Mathematics.

Project 2.

Stage 1: Problem Identification.

1.1 Description of the Problem.

A farmer wants to enclose a paddock using a fence that has a total length of 3000 meters. The farmer's goal is to maximize the enclosed area using the given amount of fencing material. The shape of the paddock is assumed to be rectangular. The farmer needs to find the dimensions (length and width) of the paddock that will maximize the area.

1.2 Statement of Intent.

The project aims to advise the farmer on how to use the available 3000 meters of fencing material to maximize the area of a rectangular paddock by applying differentiation.

1.3 Project Specifications/Parameters.

Shape of the paddock: The paddock will be in a shape of a rectangle.

Total fence length (Perimeter): The total length of the fence is fixed at 3000 meters, which will be used to enclose the paddock.

Stage 2: Investigation of Related Ideas.

Idea 1: Optimization in Agriculture.

Farmers often face the challenge of maximizing land usage with a fixed amount of resources. Optimization techniques like differentiation are commonly used in design efficient farming layouts.

Idea 2: Use of rectangular fencing.

Rectangular enclosures are commonly used in farming for paddocks, gardens and other areas. The shape is simple to construct.

Idea 3: The use of Calculus in Optimization

Using calculus to solve optimization problem is a ~~standard~~ standard approach in various fields, including agriculture.

Analysis of Ideas

Merits

Idea 1: Using optimization techniques will help farmer make the most efficient use of the available fencing material.

Idea 2: Rectangular enclosures are easier to Construct.

Idea 3: Calculus ~~require~~ (Applying differentiation) to maximize the area provides an exact solution.

Demerits.

Idea 1: The approach assumes that the terrain and land features allow for a perfect rectangular layout, which may not always be feasible in all environments.

Idea 2: While the rectangle is practical, it might not be the most efficient shape in every situation, especially if the farmer needs to enclose an area with irregular boundaries.

Idea 3: Calculus requires some understanding and may not be easily accessible to all farmers, especially if they are not familiar with mathematical techniques.

Stage 3: Refinements of chosen Idea.

3.1 Indication of choice.

The chosen solution is to maximize the

the area of the paddock using a rectangular shape with fixed total perimeter of 3000m

3.2 Justification of choice.

1. Practically : A rectangular paddock is a simple and cost effective shape for farmers to implement, making it ideal for fencing large areas.
2. Mathematical Optimization: By applying differentiation, we can precisely calculate the dimensions that result in the largest area, ensuring the farmer maximizes their land usage.

~~3.3~~

Stage 4 : Presentation of

Stage 4 : Development

Stage 4: Selecting the most suitable solution.

After comparing various mathematical methods and shapes such as circles, rectangles and square for maximising area with a fixed perimeter, the method of differentiation was chosen.

In this project, we used differentiation to show that for a given perimeter, a square provides the maximum area. This method was chosen because.

- It gives a precise mathematical proof.
- It can ~~give~~ be used for similar optimisation problem.
- It's efficient and aligns with curriculum learning goal.

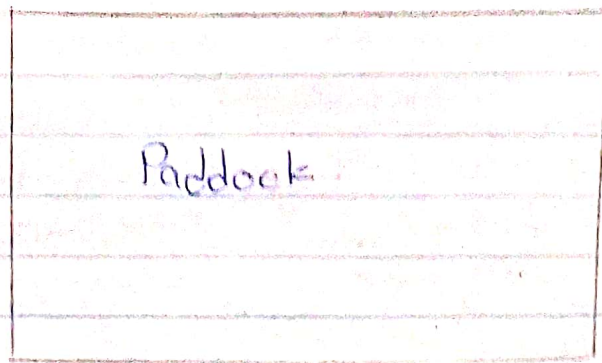
Stage 5: Refinement of selected Solution.

The selected method was tested with other examples and adjusted for accuracy. Calculations were reviewed for errors, and second derivatives were used to confirm whether it is maximum or minimum.

A clear process was followed.

- 1) Express area in terms of one variable
- 2) Differentiate and find critical points
- 3) Use second derivative to confirm maximum.
- 4) Interpret the results with real-world meaning (eg. fencing).

Stage 6: Presentation of the final solution



width.

Length

formulating the problem mathematically.

Let L = length of the paddock in metres

w = width of the paddock in metres

the perimeter of the rectangle is fixed at 3000 metres.

$$2L + 2w = 3000$$

Step 1: Expressing one variable in terms of the other.

Rearranging the perimeter formula.

$$L + w = 1500 \Rightarrow w = 1500 - L$$

Step 2: Area formula.

$$A = L \times w$$

$$A = L \times (1500 - L) = 1500L - L^2.$$

3) Applying differentiation to maximize area.

Now, have the function of Area in terms of one variable.

$$A(L) = 1500L - L^2.$$

$$\frac{dA}{dL} = 1500 - 2L.$$

Step 4 Find critical point.

Set the derivative equal to zero to find the stationary point.

$$1500 - 2L = 0 \Rightarrow L = 750.$$

Step 5 : Confirm it's maximum.

Differentiate again to find second derivative.

$$\frac{d^2A}{dL^2} = -2.$$

Since the second derivative is negative, the function has a maximum at $L = 750\text{m}$.

Stage 7 : Evaluation and recommendations.

Relevance to statement of Intent.

This project directly addresses the objective of maximizing the paddock area with a fixed fencing material. The use of differentiation provides the farmer with an optimal solution for the dimensions of the paddock.

6.2 Challenges encountered.

The main challenge was assuming that the paddock would always be rectangular, as other shapes might be more suitable in some areas. Practically, ~~pract~~ Additionally, practical considerations such as terrain and access points were not ~~off~~ factored into the mathematical model.

6.3 Recommendations

The farmer should use the dimensions 750m to maximize the enclosed area, as this yields the most efficient use of the fencing material.